

Exercise J-2

SUBJECTIVE / INTEGER TYPE QUESTIONS

BASED ON COMPOUND ANGLES

1. Prove that : $\cos^2 \alpha + \cos^2(\alpha + \beta) - 2 \cos \alpha \cdot \cos \beta \cos(\alpha + \beta) = \sin^2 \beta$

सिद्ध करो कि : $\cos^2 \alpha + \cos^2(\alpha + \beta) - 2 \cos \alpha \cdot \cos \beta \cos(\alpha + \beta) = \sin^2 \beta$

Ans.

Sol. L.H.S. =

$$\begin{aligned} & \cos^2 \alpha + \cos^2(\alpha + \beta) - 2 \cos \alpha \cos \beta \cos(\alpha + \beta) \\ &= \cos^2 \alpha + \cos(\alpha + \beta) [\cos(\alpha + \beta) - 2 \cos \alpha \cos \beta] \\ &= \cos^2 \alpha + \cos(\alpha + \beta) (\cos \alpha \cos \beta - \sin \alpha \sin \beta - 2 \cos \alpha \cos \beta) \\ &= \cos^2 \alpha - \cos(\alpha + \beta) \cos(\alpha - \beta) \\ &= \cos^2 \alpha - (\cos^2 \alpha - \sin^2 \beta) \\ &= \sin^2 \beta \quad \left\{ \because \cos^2 \alpha - \sin^2 \beta = \cos(\alpha + \beta) \cos(\alpha - \beta) \right\} \\ &= \text{R.H.S.} \end{aligned}$$

2. Prove that : $\tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha = \cot \alpha$.

सिद्ध करो कि : $\tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha = \cot \alpha$.

Ans.

Sol. L.H.S.

$$\begin{aligned} &= \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha \\ &= \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + \frac{8}{\tan 8\alpha} \\ &= \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + \frac{8(1 - \tan^2 4\alpha)}{2 \tan 4\alpha} \\ &= \tan \alpha + 2 \tan 2\alpha + \frac{8 \tan^2 4\alpha + 8 - 8 \tan^2 4\alpha}{2 \tan 4\alpha} \\ &= \tan \alpha + 2 \tan 2\alpha + \frac{4}{\tan 4\alpha} \end{aligned}$$

$$\begin{aligned}
&= \tan \alpha + 2 \tan 2\alpha + \frac{4(1 - \tan^2 2\alpha)}{2 \tan 2\alpha} \\
&= \tan \alpha + \frac{2}{\tan 2\alpha} \\
&= \frac{2 \tan^2 \alpha + 2(1 - \tan^2 \alpha)}{2 \tan \alpha} = \frac{1}{\tan \alpha} = \cot \alpha = \text{R.H.S.}
\end{aligned}$$

3. If $X = \sin\left(\theta + \frac{7\pi}{12}\right) + \sin\left(\theta - \frac{\pi}{12}\right) + \sin\left(\theta + \frac{3\pi}{12}\right)$, $Y = \cos\left(\theta + \frac{7\pi}{12}\right) + \cos\left(\theta - \frac{\pi}{12}\right) + \cos\left(\theta + \frac{3\pi}{12}\right)$ then prove that $\frac{X}{Y} - \frac{Y}{X} = 2 \tan 2\theta$.

यदि $X = \sin\left(\theta + \frac{7\pi}{12}\right) + \sin\left(\theta - \frac{\pi}{12}\right) + \sin\left(\theta + \frac{3\pi}{12}\right)$, $Y = \cos\left(\theta + \frac{7\pi}{12}\right) + \cos\left(\theta - \frac{\pi}{12}\right) + \cos\left(\theta + \frac{3\pi}{12}\right)$ तो सिद्ध कीजिए कि $\frac{X}{Y} - \frac{Y}{X} = 2 \tan 2\theta$.

Ans.

Sol. $X = \sin\left(\theta + \frac{7\pi}{12}\right) + \sin\left(\theta - \frac{\pi}{12}\right) + \sin\left(\theta + \frac{3\pi}{12}\right)$

$$\Rightarrow X = 2 \sin\left(\frac{2\theta + \frac{\pi}{2}}{2}\right) \cos\left(\frac{\pi}{3}\right) + \sin\left(\theta + \frac{\pi}{4}\right)$$

$$\Rightarrow X = \sin\left(\theta + \frac{\pi}{4}\right) \left(2 \cos \frac{\pi}{3} + 1\right)$$

$$\Rightarrow X = 2 \sin\left(\theta + \frac{\pi}{4}\right)$$

Now

$$Y = \cos\left(\theta + \frac{7\pi}{12}\right) + \cos\left(\theta - \frac{\pi}{12}\right) + \cos\left(\theta + \frac{3\pi}{12}\right)$$

$$\Rightarrow Y = 2 \cos\left(\theta + \frac{\pi}{4}\right) \cos\left(\frac{\pi}{3}\right) + \cos\left(\theta + \frac{\pi}{4}\right)$$

$$\Rightarrow Y = 2 \cos\left(\theta + \frac{\pi}{4}\right)$$

$$\text{Now } \frac{X}{Y} - \frac{Y}{X} = \frac{2 \sin\left(\theta + \frac{\pi}{4}\right)}{2 \cos\left(\theta + \frac{\pi}{4}\right)} - \frac{2 \cos\left(\theta + \frac{\pi}{4}\right)}{2 \sin\left(\theta + \frac{\pi}{4}\right)}$$

$$= \tan\left(\theta + \frac{\pi}{4}\right) - \cot\left(\theta + \frac{\pi}{4}\right)$$

$$= -\left[\cot\left(\theta + \frac{\pi}{4}\right) - \tan\left(\theta + \frac{\pi}{4}\right)\right]$$

$$= -\left[2 \cot\left(2\theta + \frac{2\pi}{4}\right)\right]$$

$$= -2 \cot\left(\frac{\pi}{2} + 2\theta\right) = 2 \tan 2\theta$$

4. Find the positive integers p, q, r, s satisfying $\tan \frac{\pi}{24} = (\sqrt{p} - \sqrt{q})(\sqrt{r} - s)$.

धनात्मक पूर्णांक p, q, r, तथा s को ज्ञात करो यदि $\tan \frac{\pi}{24} = (\sqrt{p} - \sqrt{q})(\sqrt{r} - s)$

Ans. p = 3, q = 2; r = 2; s = 1

Sol. $\because \tan^2 A = \frac{1 - \cos 2A}{1 + \cos 2A}$

$$\Rightarrow \tan^2\left(\frac{\pi}{24}\right) = \frac{1 - \cos\left(\frac{\pi}{12}\right)}{1 + \cos\left(\frac{\pi}{12}\right)} = \frac{1 - \left(\frac{\sqrt{3} + 1}{2\sqrt{2}}\right)}{1 + \left(\frac{\sqrt{3} + 1}{2\sqrt{2}}\right)}$$

$$= \left(\frac{2\sqrt{2} - (\sqrt{3} + 1)}{2\sqrt{2} + (\sqrt{3} + 1)}\right) \left(\frac{2\sqrt{2} - (\sqrt{3} + 1)}{2\sqrt{2} - (\sqrt{3} + 1)}\right)$$

$$= \frac{(2\sqrt{2} - [\sqrt{3} + 1])^2}{8 - 4 - 2\sqrt{3}} = \frac{(2\sqrt{2} - \sqrt{3} - 1)^2}{4 - 2\sqrt{3}}$$

$$\begin{aligned}
&= \frac{(4+2\sqrt{3})(2\sqrt{2}-\sqrt{3}-1)^2}{(4+2\sqrt{3})(4-2\sqrt{3})} \\
&= \frac{(4+2\sqrt{3})[8+4+2\sqrt{3}-4\sqrt{2}(\sqrt{3}+1)]}{4} \\
&= \frac{(4+2\sqrt{3})[12+2\sqrt{3}-4\sqrt{6}-4\sqrt{2}]}{4} \\
&= \left[\frac{48+8\sqrt{3}-16\sqrt{6}-16\sqrt{2}+24\sqrt{3}+12-24\sqrt{2}-8\sqrt{6}}{4} \right] \\
&= 15-6\sqrt{6}+8\sqrt{3}-10\sqrt{2} \\
&= (5-2\sqrt{6})(3-2\sqrt{2}) \\
&= (\sqrt{3}-\sqrt{2})^2 (\sqrt{2}-1)^2 \\
&\Rightarrow \tan\left(\frac{\pi}{24}\right) = (\sqrt{3}-\sqrt{2})(\sqrt{2}-1) \\
&\Rightarrow p=3, q=2, r=2, s=1
\end{aligned}$$

5. If the value of the expression $\sin 25^\circ \cdot \sin 35^\circ \cdot \sin 85^\circ$ can be expressed as $\frac{\sqrt{a} + \sqrt{b}}{c}$ where $a, b, c \in \mathbb{N}$ and are in their lowest form, find the value of $(a+b+c)$.

यदि व्यंजक $\sin 25^\circ \cdot \sin 35^\circ \cdot \sin 85^\circ$ को $\frac{\sqrt{a} + \sqrt{b}}{c}$ के रूप में व्यक्त किया जाता है जहाँ $a, b, c \in \mathbb{N}$ (न्यूनतम रूप) हो तो $a+b+c$ का मान बताओं।

Ans. 24

Sol.

$$\begin{aligned}
&\sin(25^\circ) \cdot \sin(35^\circ) \cdot \sin(85^\circ) \\
&= \sin(25^\circ) \sin(60^\circ - 25^\circ) \sin(60^\circ + 25^\circ) \\
&= \frac{1}{4} \sin(75^\circ) \\
&= \frac{1}{4} \cos(15^\circ) = \frac{\sqrt{3}+1}{8\sqrt{2}} = \frac{\sqrt{\frac{3}{2}} + \sqrt{\frac{1}{2}}}{8}
\end{aligned}$$

$$= \frac{\sqrt{6} + \sqrt{2}}{16}$$

$$\Rightarrow a = 6, b = 2, c = 16$$

$$\Rightarrow a + b + c = 24$$

6. Prove that :

सिद्ध करो कि :

$$(a) \tan 20^\circ \cdot \tan 40^\circ \cdot \tan 60^\circ \cdot \tan 80^\circ = 3$$

$$(b) \tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ = 4.$$

$$(c) \sin^4 \frac{\pi}{16} + \sin^4 \frac{3\pi}{16} + \sin^4 \frac{5\pi}{16} + \sin^4 \frac{7\pi}{16} = \frac{3}{2}$$

Ans.

Sol. (a) $\tan 20^\circ \cdot \tan 40^\circ \cdot \tan 60^\circ \cdot \tan 80^\circ$

$$= \tan 20^\circ \cdot \tan (60^\circ - 40^\circ) \cdot \tan (60^\circ + 20^\circ) \cdot \sqrt{3}$$

$$= \tan (3 \cdot 20^\circ) \cdot \sqrt{3} = 3$$

$$(b) \tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$$

$$= \tan 9^\circ + \tan 81^\circ - (\tan 27^\circ + \tan 63^\circ)$$

$$= \tan 9^\circ + \cot 9^\circ - (\tan 27^\circ + \cot 27^\circ)$$

$$= 2 \operatorname{cosec} (18^\circ) - 2 \operatorname{cosec} (54^\circ)$$

$$= \frac{2}{\sin (18^\circ)} - \frac{2}{\sin 54^\circ} = \frac{2}{\sin 18^\circ} - \frac{2}{\cos 36^\circ} = \frac{2}{\left(\frac{\sqrt{5}-1}{4}\right)} - \frac{2}{\left(\frac{\sqrt{5}+1}{4}\right)} = 4$$

$$(C) \sin^4 \left(\frac{\pi}{16}\right) + \sin^4 \left(\frac{3\pi}{16}\right) + \sin^4 \left(\frac{5\pi}{16}\right) + \sin^4 \left(\frac{7\pi}{16}\right)$$

$$= \sin^4 \left(\frac{\pi}{16}\right) + \sin^4 \left(\frac{3\pi}{16}\right) + \sin^4 \left(\frac{\pi}{2} - \frac{3\pi}{16}\right) + \sin^4 \left(\frac{\pi}{2} - \frac{\pi}{16}\right)$$

$$= \sin^4 \left(\frac{\pi}{16}\right) + \cos^4 \left(\frac{\pi}{16}\right) + \sin^4 \left(\frac{3\pi}{16}\right) + \cos^4 \left(\frac{3\pi}{16}\right)$$

$$= \left[\sin^2 \left(\frac{\pi}{16}\right) + \cos^2 \left(\frac{\pi}{16}\right) \right]^2 + \left[\sin^2 \left(\frac{3\pi}{16}\right) + \cos^2 \left(\frac{3\pi}{16}\right) \right]^2 - 2 \left(\sin^2 \frac{\pi}{16} \cos^2 \frac{\pi}{16} + \sin^2 \frac{3\pi}{16} \cos^2 \frac{3\pi}{16} \right)$$

$$= 2 - \frac{1}{2} \left(\sin^2 \frac{\pi}{8} + \sin^2 \left(\frac{\pi}{2} - \frac{\pi}{8}\right) \right)$$

$$= 2 - \frac{1}{2} = \frac{3}{2}$$

7. Calculate without using trigonometric tables : $4 \cos 20^\circ - \sqrt{3} \cot 20^\circ$

बिना त्रिकोणमितीय सारणी के निम्नलिखित को सरल करो : $4 \cos 20^\circ - \sqrt{3} \cot 20^\circ$

Ans. -1

Sol.

$$\begin{aligned} & 4 \cos 20^\circ - \sqrt{3} \cot 20^\circ \\ &= \frac{4 \cos 20^\circ \sin 20^\circ - \sqrt{3} \cos 20^\circ}{\sin 20^\circ} \\ &= \frac{1}{\sin 20^\circ} \cdot 2 \left[\sin 40^\circ - \frac{\sqrt{3}}{2} \cos 20^\circ \right] \\ &= \frac{1}{\sin 20^\circ} [2 \sin 40^\circ - 2 \sin 60^\circ \cos 20^\circ] \\ &= \frac{2 \sin 40^\circ - \sin(80^\circ) - \sin(40^\circ)}{\sin 20^\circ} \\ &= \frac{2 \cos(60^\circ) \sin(-20^\circ)}{\sin 20^\circ} = -1 \end{aligned}$$

8. Calculate without using trigonometric tables : $\frac{2 \cos 40^\circ - \cos 20^\circ}{\sin 20^\circ}$

बिना त्रिकोणमितीय सारणी के निम्नलिखित को सरल करो : $\frac{2 \cos 40^\circ - \cos 20^\circ}{\sin 20^\circ}$

Ans. $\sqrt{3}$

Sol.

$$\begin{aligned} & \frac{2 \cos 40^\circ - \cos 20^\circ}{\sin 20^\circ} = \frac{2}{\sin 20^\circ} \left[\cos 40^\circ - \frac{1}{2} \cos 20^\circ \right] \\ &= \frac{2 \cos 40^\circ - 2 \cos 60^\circ \cos 20^\circ}{\sin 20^\circ} \\ &= \frac{2 \cos 40^\circ - \cos 80^\circ - \cos 40^\circ}{\sin 20^\circ} = \frac{2 \sin 60^\circ \sin 20^\circ}{\sin 20^\circ} \\ &= \sqrt{3} \end{aligned}$$

9. Calculate without using trigonometric tables : $\cos^6 \frac{\pi}{16} + \cos^6 \frac{3\pi}{16} + \cos^6 \frac{5\pi}{16} + \cos^6 \frac{7\pi}{16}$

बिना त्रिकोणमितीय सारणी के निम्नलिखित को सरल करो : $\cos^6 \frac{\pi}{16} + \cos^6 \frac{3\pi}{16} + \cos^6 \frac{5\pi}{16} + \cos^6 \frac{7\pi}{16}$

Ans. $\frac{5}{4}$

Sol.

$$\begin{aligned} & \cos^6 \frac{\pi}{16} + \cos^6 \frac{3\pi}{16} + \cos^6 \frac{5\pi}{16} + \cos^6 \frac{7\pi}{16} \\ &= \cos^6 \frac{\pi}{16} + \cos^6 \frac{3\pi}{16} + \cos^6 \left(\frac{\pi}{2} - \frac{3\pi}{16} \right) + \cos^6 \left(\frac{\pi}{2} - \frac{7\pi}{16} \right) \\ &= \cos^6 \frac{\pi}{16} + \sin^6 \frac{\pi}{16} + \cos^6 \frac{3\pi}{16} + \sin^6 \frac{3\pi}{16} \\ &= \left(\sin^2 \frac{\pi}{16} + \cos^2 \frac{\pi}{16} \right)^3 - 3 \sin^2 \frac{\pi}{16} \cos^2 \frac{\pi}{16} \left(\sin^2 \frac{\pi}{16} + \cos^2 \frac{\pi}{16} \right) \\ & \quad + \left(\sin^2 \frac{3\pi}{16} + \cos^2 \frac{3\pi}{16} \right)^3 - 3 \sin^2 \frac{3\pi}{16} \cos^2 \frac{3\pi}{16} \left(\sin^2 \frac{3\pi}{16} + \cos^2 \frac{3\pi}{16} \right) \\ &= 2 - 3 \left(\sin^2 \frac{\pi}{16} \cos^2 \frac{\pi}{16} \right) - 3 \sin^2 \frac{3\pi}{16} \cos^2 \frac{3\pi}{16} \\ &= 2 - \frac{3}{4} \left(\sin^2 \frac{\pi}{8} \right) - \frac{3}{4} \left(\sin^2 \frac{3\pi}{8} \right) \\ &= 2 - \frac{3}{8} \left(1 - \cos \frac{\pi}{4} \right) - \frac{3}{8} \left(1 - \cos \frac{3\pi}{4} \right) \\ &= 2 - \frac{3}{4} + \frac{3}{8} \left(\cos \frac{\pi}{4} + \cos \frac{3\pi}{4} \right) \\ &= 2 - \frac{3}{4} + \frac{3}{8} \left(\cos \frac{\pi}{4} - \cos \frac{\pi}{4} \right) \\ &= \frac{5}{4} \end{aligned}$$

10. If α , β and γ are in A.P., show that $\cot \beta = \frac{\sin \alpha - \sin \gamma}{\cos \gamma - \cos \alpha}$

यदि α, β तथा γ समान्तर श्रेणी में हो, तो सिद्ध कीजिए कि $\cot \beta = \frac{\sin \alpha - \sin \gamma}{\cos \gamma - \cos \alpha}$

Ans.

Sol. Given $\alpha, \beta, \gamma \rightarrow$ A.P.

$$\Rightarrow 2\beta = \alpha + \gamma$$

R.H.S.

$$\begin{aligned} &= \frac{\sin \alpha - \sin \gamma}{\cos \gamma - \cos \alpha} = \frac{2 \cos \left(\frac{\alpha + \gamma}{2} \right) \sin \left(\frac{\alpha - \gamma}{2} \right)}{2 \sin \left(\frac{\alpha + \gamma}{2} \right) \sin \left(\frac{\alpha - \gamma}{2} \right)} \\ &= \cot \left(\frac{\alpha + \gamma}{2} \right) = \cot(\beta) \end{aligned}$$

11. If $\tan \beta = \frac{\tan \alpha + \tan \gamma}{1 + \tan \alpha \tan \gamma}$, prove that $\sin 2\beta = \frac{\sin 2\alpha + \sin 2\gamma}{1 + \sin 2\alpha \sin 2\gamma}$

यदि $\tan \beta = \frac{\tan \alpha + \tan \gamma}{1 + \tan \alpha \tan \gamma}$ हो, तो सिद्ध कीजिए कि $\sin 2\beta = \frac{\sin 2\alpha + \sin 2\gamma}{1 + \sin 2\alpha \sin 2\gamma}$

Ans.

Sol. Given

$$\tan \beta = \frac{\tan \alpha + \tan \gamma}{1 + \tan \alpha \tan \gamma}$$

$$\Rightarrow \tan \beta = \frac{\sin(\alpha + \gamma)}{\cos(\alpha - \gamma)}$$

$$\Rightarrow \sin 2\beta = \frac{2 \tan \beta}{1 + \tan^2 \beta} = \frac{2 \sin(\alpha + \gamma) \cdot \cos(\alpha - \gamma)}{\cos^2(\alpha - \gamma) + \sin^2(\alpha + \gamma)}$$

$$\Rightarrow \sin 2\beta = \frac{\sin 2\alpha + \sin 2\gamma}{(\cos \alpha \cos \gamma + \sin \alpha \sin \gamma)^2 + (\sin \alpha \cos \gamma + \cos \alpha \sin \gamma)^2}$$

$$\Rightarrow \sin 2\beta = \frac{\sin 2\alpha + \sin 2\gamma}{\cos^2 \gamma + \sin^2 \gamma + \sin 2\alpha \sin 2\gamma} = \frac{\sin 2\alpha + \sin 2\gamma}{1 + \sin 2\alpha \sin 2\gamma}$$

COMPONENDO AND DIVIDENDO

12. If $m \tan(\theta - 30^\circ) = n \tan(\theta + 120^\circ)$, show that $\cos 2\theta = \frac{m+n}{2(m-n)}$.

यदि $m \tan(\theta - 30^\circ) = n \tan(\theta + 120^\circ)$, तो सिद्ध करो कि $\cos 2\theta = \frac{m+n}{2(m-n)}$

Ans.

Sol. $m \tan(\theta - 30^\circ) = n \tan(\theta + 120^\circ)$

$$\Rightarrow \frac{m}{n} = \frac{\tan(\theta + 120^\circ)}{\tan(\theta - 30^\circ)} = \frac{\sin(\theta + 120^\circ) \cos(\theta - 30^\circ)}{\sin(\theta - 30^\circ) \cos(\theta + 120^\circ)}$$

By C & D Rule

$$\Rightarrow \frac{m+n}{m-n} = \frac{\sin(\theta + 120^\circ + \theta - 30^\circ)}{\sin(\theta + 120^\circ - \theta + 30^\circ)} = \frac{\sin(90^\circ + 2\theta)}{\sin(150^\circ)}$$

$$\Rightarrow \frac{m+n}{m-n} = \frac{\cos(2\theta)}{\cos(60^\circ)} \Rightarrow \cos(2\theta) = \frac{m+n}{2(m-n)}$$

CONTINUED PRODUCT

13. Let $P(k) = \left(1 + \cos \frac{\pi}{4k}\right) \left(1 + \cos \frac{(2k-1)\pi}{4k}\right) \left(1 + \cos \frac{(2k+1)\pi}{4k}\right) \left(1 + \cos \frac{(4k-1)\pi}{4k}\right)$ then find the value of (a) $P(5)$ and (b) $P(6)$.

माना कि $P(k) = \left(1 + \cos \frac{\pi}{4k}\right) \left(1 + \cos \frac{(2k-1)\pi}{4k}\right) \left(1 + \cos \frac{(2k+1)\pi}{4k}\right) \left(1 + \cos \frac{(4k-1)\pi}{4k}\right)$ तो (a) $P(5)$ तथा (b) $P(6)$ का मान बताइये।

Ans. (a) $\frac{3-\sqrt{5}}{32}$; (b) $\frac{2-\sqrt{3}}{16}$

Sol. $P(k) = \left(1 + \cos \frac{\pi}{4k}\right) \left(1 + \cos \frac{(2k-1)\pi}{4k}\right) \left(1 + \cos \frac{(2k+1)\pi}{4k}\right) \left(1 + \cos \frac{(4k-1)\pi}{4k}\right)$

$$= \left(1 + \cos \frac{\pi}{4k}\right) \left(1 + \cos \left(\frac{\pi}{2} - \frac{\pi}{4k}\right)\right) \left(1 + \cos \left(\frac{\pi}{2} + \frac{\pi}{4k}\right)\right) \left(1 + \cos \left(\pi - \frac{\pi}{4k}\right)\right)$$

$$\begin{aligned}
&= \left(1 + \cos \frac{\pi}{4k}\right) \left(1 - \cos \frac{\pi}{4k}\right) \left(1 + \sin \left(\frac{\pi}{4k}\right)\right) \left(1 - \sin \left(\frac{\pi}{4k}\right)\right) \\
&= \left(1 - \cos^2 \left(\frac{\pi}{4k}\right)\right) \left(1 - \sin^2 \left(\frac{\pi}{4k}\right)\right) \\
&= \sin^2 \frac{\pi}{4k} \cdot \cos^2 \frac{\pi}{4k} \\
&= \frac{1}{4} \left(\sin \frac{\pi}{2k}\right)^2
\end{aligned}$$

$$\begin{aligned}
\text{(a) } P(5) &= \frac{1}{4} \left(\sin \frac{\pi}{10}\right)^2 = \frac{1}{4} (\sin 18^\circ)^2 \\
&= \frac{1}{4} \left(\frac{\sqrt{5}-1}{4}\right)^2 = \frac{1}{64} (6-2\sqrt{5}) \\
&= \frac{1}{32} (3-\sqrt{5})
\end{aligned}$$

$$\begin{aligned}
\text{(b) } P(6) &= \frac{1}{4} \left[\sin \left(\frac{\pi}{12}\right)\right]^2 \\
&= \frac{1}{4} [\sin 15^\circ]^2 = \frac{1}{4} \left[\frac{\sqrt{3}-1}{2\sqrt{2}}\right]^2 \\
&= \frac{2-\sqrt{3}}{16}
\end{aligned}$$

14. Given that $(1 + \tan 1^\circ)(1 + \tan 2^\circ) \dots (1 + \tan 45^\circ) = 2^n$, find n .

दिया है कि $(1 + \tan 1^\circ)(1 + \tan 2^\circ) \dots (1 + \tan 45^\circ) = 2^n$ तो n का मान बताओं।

Ans. $n = 23$

Sol. $(1 + \tan 1^\circ)(1 + \tan 2^\circ) \dots (1 + \tan 44^\circ)(1 + \tan 45^\circ) = 2^n$

$$\because (1 + \tan A)(1 + \tan B) = 2 \quad \text{If } A + B = 45^\circ$$

$$\Rightarrow 2^{22} \cdot (1 + \tan 45^\circ) = 2^n$$

$$\Rightarrow n = 23$$

15. The value of $\sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14} \sin \frac{7\pi}{14} \sin \frac{9\pi}{14} \sin \frac{11\pi}{14} \sin \frac{13\pi}{14}$ is equal to _____.

$\sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14} \sin \frac{7\pi}{14} \sin \frac{9\pi}{14} \sin \frac{11\pi}{14} \sin \frac{13\pi}{14}$ का मान बराबर होगा _____

Ans. $\frac{1}{64}$

Sol.

$$\begin{aligned} & \sin\left(\frac{\pi}{14}\right) \sin\left(\frac{3\pi}{14}\right) \sin\left(\frac{5\pi}{14}\right) \sin\left(\frac{7\pi}{14}\right) \sin\left(\frac{9\pi}{14}\right) \sin\left(\frac{11\pi}{14}\right) \sin\left(\frac{13\pi}{14}\right) \\ &= \sin\left(\frac{\pi}{14}\right) \sin\left(\frac{3\pi}{14}\right) \sin\left(\frac{5\pi}{14}\right) \sin\left(\frac{\pi}{2}\right) \sin\left(\pi - \frac{5\pi}{14}\right) \sin\left(\pi - \frac{3\pi}{14}\right) \sin\left(\pi - \frac{\pi}{14}\right) \\ &= \sin^2\left(\frac{\pi}{14}\right) \sin^2\left(\frac{3\pi}{14}\right) \sin^2\left(\frac{5\pi}{14}\right) \\ &= \left[\sin\left(\frac{\pi}{2} - \frac{3\pi}{7}\right) \sin\left(\frac{\pi}{2} - \frac{2\pi}{7}\right) \sin\left(\frac{\pi}{2} - \frac{\pi}{7}\right) \right]^2 \\ &= \left[\cos \frac{3\pi}{7} \cos \frac{2\pi}{7} \cos \frac{\pi}{7} \right]^2 \\ &= \left[\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \right]^2 \\ &= \left[\frac{\sin\left(\frac{8\pi}{7}\right)}{8 \cdot \sin\left(\frac{\pi}{7}\right)} \right]^2 = \left[\frac{\sin\left(\pi + \frac{\pi}{7}\right)}{8 \cdot \sin\left(\frac{\pi}{7}\right)} \right]^2 = \frac{1}{64} \end{aligned}$$

16. $\sin(B + 2C) + \sin(C + 2A) + \sin(A + 2B) = 4 \sin \frac{B-C}{2} \sin \frac{C-A}{2} \sin \frac{A-B}{2}$

Ans.

Sol.

$$\begin{aligned} & \sin(B + 2C) + \sin(C + 2A) + \sin(A + 2B) \\ &= \sin(B + C + C) + \sin(C + A + A) + \sin(A + B + B) \\ &= \sin(\pi - A + C) \sin(\pi - B + A) + \sin(\pi - C + B) \\ &= \sin(A - C) + \sin(B - A) + \sin(C - B) \\ &= 2 \sin \frac{B-C}{2} \cos \frac{2A-B-C}{2} + 2 \sin \frac{C-B}{2} \cos \frac{C-B}{2} \\ &= 2 \sin \frac{B-C}{2} \left\{ \cos \frac{2A-B-C}{2} - \cos \frac{B-C}{2} \right\} \end{aligned}$$

$$= 4 \sin \frac{B-C}{2} \sin \frac{A-C}{2} \sin \frac{B-A}{2}$$

$$= 4 \sin \frac{B-C}{2} \sin \frac{C-A}{2} \sin \frac{A-B}{2}$$

17. $\cos \frac{A}{2} + \cos \frac{B}{2} - \cos \frac{C}{2} = 4 \cos \frac{\pi+A}{4} \cos \frac{\pi+B}{4} \cos \frac{\pi-C}{4}$

Ans.

Sol. $4 \cos \frac{\pi+A}{4} \cos \frac{\pi+B}{4} \cos \frac{\pi-C}{4} = 2 \cos \frac{\pi-C}{4} \left\{ \cos \frac{2\pi+A+B}{4} + \cos \frac{A-B}{4} \right\}$

$$= 2 \cos \frac{A+B}{4} \left\{ \cos \left(\frac{\pi}{2} + \frac{A+B}{4} \right) + \cos \left(\frac{A-B}{4} \right) \right\}$$

$$= -2 \cos \frac{A+B}{4} \sin \frac{A+B}{4} + 2 \cos \frac{A+B}{4} \cos \frac{A-B}{4}$$

$$= -\sin \frac{A+B}{2} + \cos \frac{A}{2} + \cos \frac{B}{2}$$

$$= \cos \frac{A}{2} + \cos \frac{B}{2} - \cos \frac{C}{2}$$

RANGE OF TRIGONOMETRIC FUNCTIONS

18. If $y = 10 \cos^2 x - 6 \sin x \cos x + 2 \sin^2 x$, then find the greatest and least value of y .

यदि $y = 10 \cos^2 x - 6 \sin x \cos x + 2 \sin^2 x$, तो y का अधिकतम तथा न्यूनतम मान बताओं।

Ans. $y_{\max} = 11, y_{\min} = 1$

Sol. $y = 10 \cos^2 x - 6 \sin x \cos x + 2 \sin^2 x$

$$= 10 \left(\frac{1 + \cos 2x}{2} \right) - 3 \sin 2x + 2 \left(\frac{1 - \cos 2x}{2} \right)$$

$$= 5 + 5 \cos 2x - 3 \sin 2x + 1 - \cos 2x$$

$$= 6 + 4 \cos 2x - 3 \sin 2x$$

Range of $4 \cos 2x - 3 \sin 2x$ is

$$\left[-\sqrt{16+9}, \sqrt{16+9} \right]$$

$$\Rightarrow y \in [1, 11]$$

$$\Rightarrow y_{\max} = 11 \quad y_{\min} = 1$$

19. If $y = 1 + 2\sin x + 3\cos^2 x$, find the maximum and minimum values of $y \forall x \in \mathbb{R}$.

यदि $y = 1 + 2\sin x + 3\cos^2 x$ तो x के प्रत्येक मान के लिये y का अधिकतम तथा न्यूनतम मान बताओं।

Ans. $y_{\max} = \frac{13}{3}, y_{\min} = -1$

Sol. $y = 1 + 2\sin x + 3\cos^2 x$
 $= 1 + 2\sin x + 3(1 - \sin^2 x)$
 $= 4 + 2\sin x - 3\sin^2 x$

$$= -3 \left[\sin^2 x - \frac{2}{3} \sin x - \frac{4}{3} \right]$$

$$= -3 \left[\left(\sin x - \frac{1}{3} \right)^2 - \frac{1}{9} - \frac{4}{3} \right]$$

$$= \frac{1}{3} + 4 - 3 \left[\left(\sin x - \frac{1}{3} \right)^2 \right] = \frac{13}{3} - 3 \left[\left(\sin x - \frac{1}{3} \right)^2 \right]$$

$$\because -1 \leq \sin x \leq +1$$

$$\Rightarrow -\frac{4}{3} \leq \sin x - \frac{1}{3} \leq \frac{2}{3}$$

$$\Rightarrow 0 \leq \left(\sin x - \frac{1}{3} \right)^2 \leq \frac{16}{9}$$

$$\Rightarrow -\frac{16}{3} \leq -3 \left(\sin x - \frac{1}{3} \right)^2 \leq 0$$

$$\Rightarrow \frac{-3}{3} \leq \frac{13}{3} - 3 \left(\sin x - \frac{1}{3} \right)^2 \leq \frac{13}{3}$$

$$\Rightarrow y \in \left[-1, \frac{13}{3} \right]$$

$$\Rightarrow y_{\max} = \frac{13}{3} \quad y_{\min} = -1$$

20. If $y = 9\sec^2 x + 16\operatorname{cosec}^2 x$, find the minimum value of $y \forall x \in \mathbb{R}$.

यदि $y = 9\sec^2 x + 16\operatorname{cosec}^2 x$, तो x के सभी मान के लिये y का न्यूनतम मान बताओं।

Ans. 49

Sol. $y = 9\sec^2 x + 16\operatorname{cosec}^2 x$

$$\begin{aligned}
y &= 9(1 + \tan^2 x) + 16(1 + \cot^2 x) \\
&= 25 + 9 \tan^2 x + 16 \cot^2 x \\
\therefore a^2 \tan^2 x + b^2 \cot^2 x &\in [2ab, \infty) \\
\Rightarrow 9 \tan^2 x + 16 \cot^2 x &\in [24, \infty) \\
\Rightarrow y &\in [49, \infty) \\
\Rightarrow y_{\min} &= 49
\end{aligned}$$

21. If $a \leq 3 \cos \left(\theta + \frac{\pi}{3} \right) + 5 \cos \theta + 3 \leq b$, find a and b .

यदि $a \leq 3 \cos \left(\theta + \frac{\pi}{3} \right) + 5 \cos \theta + 3 \leq b$ तो a व b का मान बताओं।

Ans. $a = -4$ and $b = 10$

Sol. $a \leq 3 \cos \left(\theta + \frac{\pi}{3} \right) + 5 \cos \theta + 3 \leq b$

$$\Rightarrow a \leq 3 \left[\cos \theta \cos \frac{\pi}{3} - \sin \theta \sin \frac{\pi}{3} \right] + 5 \cos \theta + 3 \leq b$$

$$\Rightarrow a \leq 3 \left[\frac{\cos \theta}{2} - \frac{\sqrt{3}}{2} \sin \theta \right] + 5 \cos \theta + 3 \leq b$$

$$\Rightarrow a \leq \frac{3}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta + 5 \cos \theta + 3 \leq b$$

$$\Rightarrow a \leq \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta + 3 \leq b$$

$$\therefore \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta \in \left[-\sqrt{\frac{169}{4} + \frac{27}{4}}, \sqrt{\frac{169}{4} + \frac{27}{4}} \right]$$

$$\Rightarrow 3 + \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta \in [3 - 7, 3 + 7]$$

$$a = -4, b = 10$$

22. Show that $2^{\sin x} + 2^{\cos x} \geq 2^{1 - \frac{1}{\sqrt{2}}}$.

सिद्ध कीजिए $2^{\sin x} + 2^{\cos x} \geq 2^{1 - \frac{1}{\sqrt{2}}}$.

Ans.

Sol. Let $a = 2^{\sin x}$, $b = 2^{\cos x}$

$$\Rightarrow \frac{a+b}{2} \geq \sqrt{ab}$$

$$\Rightarrow 2^{\sin x} + 2^{\cos x} \geq 2\sqrt{2^{\sin x + \cos x}}$$

$$\left\{ \begin{array}{l} -\sqrt{2} \leq \sin x + \cos x \leq \sqrt{2} \\ \Rightarrow 2^{-\sqrt{2}} \leq 2^{\sin x + \cos x} \leq 2^{\sqrt{2}} \end{array} \right\}$$

$$\Rightarrow 2^{\sin x} + 2^{\cos x} \geq 2^{1-\frac{1}{\sqrt{2}}}$$

23. Show that $\frac{3 + \cos x}{\sin x} \forall x \in \mathbb{R}$ can not have any value between $-2\sqrt{2}$ and $2\sqrt{2}$. What inference can you draw about the values of $\frac{\sin x}{3 + \cos x}$?

सिद्ध करो कि x के प्रत्येक मान के लिये $\frac{3 + \cos x}{\sin x}$ का मान $-2\sqrt{2}$ तथा $2\sqrt{2}$ के मध्य कभी नहीं आयेगा। इस परिणाम की मदद

से आप $\frac{\sin x}{3 + \cos x}$ के मानों के बारे में क्या कह सकते हैं?

Ans. $\left[-\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}} \right]$

Sol. $\frac{3 + \cos x}{\sin x} = y \Rightarrow y \sin x - 3 = \cos x$

Square both sides

$$\Rightarrow y^2 \sin^2 x + 9 - 6y \sin x = 1 - \sin^2 x$$

$$\Rightarrow (y^2 + 1) \sin^2 x - 6y \sin x + 8 = 0$$

For real values of x ; $D \geq 0$

$$\Rightarrow 36y^2 - 32(y^2 + 1) \geq 0$$

$$\Rightarrow y^2 \geq 8$$

$$\Rightarrow y \in (-\infty, -2\sqrt{2}] \cup [2\sqrt{2}, +\infty)$$

$$\Rightarrow \frac{1}{y} \in \left[-\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}} \right]$$

24. The maximum value of the expression $\frac{1}{\sin^2 \theta + 3 \sin \theta \cos \theta + 5 \cos^2 \theta}$ is. [JEE-2010]

व्यंजक $\frac{1}{\sin^2 \theta + 3 \sin \theta \cos \theta + 5 \cos^2 \theta}$ का अधिकतम मान बताओं। [JEE-2010]

Ans. 2

Sol.
$$f(\theta) = \frac{1}{\sin^2 \theta + 3 \sin \theta \cos \theta + 5 \cos^2 \theta} = \frac{1}{\frac{1 - \cos 2\theta}{2} + \frac{3}{2} \sin 2\theta + \frac{5(1 + \cos 2\theta)}{2}} = \frac{2}{6 + 3 \sin 2\theta + 4 \cos 2\theta}$$

$$\therefore f(\theta)_{\max} = \frac{2}{6-5} = 2; \quad \left\{ \because 3 \sin 2\theta + 4 \cos 2\theta \in [-5, 5] \right\}$$

SUMMATION OF SERIES

25. If $f(\theta) = \sum_{n=1}^6 \operatorname{cosec} \left(\theta + \frac{(n-1)\pi}{4} \right) \operatorname{cosec} \left(\theta + \frac{n\pi}{4} \right)$, where $0 < \theta < \frac{\pi}{2}$, then find the minimum value of $f(\theta)$.

यदि $f(\theta) = \sum_{n=1}^6 \operatorname{cosec} \left(\theta + \frac{(n-1)\pi}{4} \right) \operatorname{cosec} \left(\theta + \frac{n\pi}{4} \right)$, जहाँ $0 < \theta < \frac{\pi}{2}$, तो $f(\theta)$ का न्यूनतम मान क्या होगा।

Ans. $2\sqrt{2}$

Sol.
$$f(\theta) = \sum_{n=1}^6 \operatorname{cosec} \left(\theta + \frac{(n-1)\pi}{4} \right) \operatorname{cosec} \left(\theta + \frac{n\pi}{4} \right)$$

$$T_n = \operatorname{cosec} \left(\theta + \frac{(n-1)\pi}{4} \right) \operatorname{cosec} \left(\theta + \frac{n\pi}{4} \right)$$

$$T_n = \sqrt{2} \frac{\sin \left[\left(\theta + \frac{n\pi}{4} \right) - \left(\theta + \frac{(n-1)\pi}{4} \right) \right]}{\left(\sin \left(\theta + \frac{(n-1)\pi}{4} \right) \sin \left(\theta + \frac{n\pi}{4} \right) \right)}$$

$$\begin{aligned}
\Rightarrow f(\theta) &= \sqrt{2} \left[\frac{\sin\left(\theta + \frac{n\pi}{4}\right) \cos\left(\theta + \frac{(n-1)\pi}{4}\right) - \cos\left(\theta + \frac{n\pi}{4}\right) \sin\left(\theta + \frac{(n-1)\pi}{4}\right)}{\sin\left(\theta + \frac{(n-1)\pi}{4}\right) \sin\left(\theta + \frac{n\pi}{4}\right)} \right] \\
&= \sqrt{2} \left[\cot\left(\theta + \frac{(n-1)\pi}{4}\right) - \cot\left(\theta + \frac{n\pi}{4}\right) \right] \\
f(\theta) &= T_1 + T_2 + T_3 + \dots + T_6 \\
&= \sqrt{2} \left[\cot(\theta) - \cot\left(\theta + \frac{\pi}{4}\right) + \cot\left(\theta + \frac{\pi}{4}\right) - \cot\left(\theta + \frac{\pi}{2}\right) + \dots + \cot\left(\theta + \frac{5\pi}{4}\right) - \cot\left(\theta + \frac{3\pi}{2}\right) \right] \\
&= \sqrt{2} [\cot \theta + \tan \theta] \\
&= \sqrt{2} \cdot 2 \operatorname{cosec} 2\theta \\
&\because 0 < 2\theta < \pi \\
&\Rightarrow \text{Minimum value of } \operatorname{cosec} 2\theta \text{ is } 1 \\
&\text{Minimum value of } f(\theta) \text{ is } 2\sqrt{2}
\end{aligned}$$

26. Let $k = 1^\circ$, then prove that $\sum_{n=0}^{88} \frac{1}{\cos nk \cdot \cos(n+1)k} = \frac{\cos k}{\sin^2 k}$.

यदि $k = 1^\circ$ तो सिद्ध करो कि $\sum_{n=0}^{88} \frac{1}{\cos nk \cdot \cos(n+1)k} = \frac{\cos k}{\sin^2 k}$

Ans.

Sol. $\sum_{n=0}^{88} \frac{1}{\cos nk \cdot \cos(n+1)k}$

Let $T_n = \frac{1}{\cos nk \cdot \cos(n+1)k} = \frac{\sin[(n+1)k - nk]}{\sin k \cos nk \cdot \cos(n+1)k}$

$\Rightarrow T_n = \frac{1}{\sin k} \left[\frac{\sin(n+1)k \cos(nk) - \cos(n+1)k \sin nk}{\cos(nk) \cdot \cos(n+1)k} \right]$

$\Rightarrow T_n = \frac{1}{\sin k} [\tan(n+1)k - \tan nk]$

$$\Rightarrow \sum_{n=0}^{88} \frac{1}{\cos(nk) \cdot \cos(n+1)k} = \frac{1}{\sin k} \left[\begin{array}{l} \tan k - \tan 0 \\ + \tan 2k - \tan k \\ + \tan 3k - \tan 2k \\ \vdots \\ + \tan 88k - \tan 87k \\ + \tan 89k - \tan 88k \end{array} \right]$$

$$= \frac{1}{\sin k} [\tan 89k] = \frac{\sin(89k)}{\cos(89k) \sin k}$$

$$= \frac{\sin(89^\circ)}{\cos(89^\circ) \sin 1^\circ} = \frac{\cos(1^\circ)}{\sin(1^\circ)^2} = \frac{\cos k}{\sin^2 k} \quad \{\because k = 1^\circ\}$$

27. Prove that : $\operatorname{cosec} \theta + \operatorname{cosec} 2\theta + \operatorname{cosec} 2^2 \theta + \dots + \operatorname{cosec} 2^{n-1} \theta = \cot(\theta/2) - \cot 2^{n-1} \theta$.

सिद्ध कीजिए : $\operatorname{cosec} \theta + \operatorname{cosec} 2\theta + \operatorname{cosec} 2^2 \theta + \dots + \operatorname{cosec} 2^{n-1} \theta = \cot(\theta/2) - \cot 2^{n-1} \theta$.

Ans.

Sol. $\operatorname{cosec} \theta + \operatorname{cosec} 2\theta + \operatorname{cosec} 2^2 \theta + \dots + \operatorname{cosec} (2^{n-1} \theta) = \cot\left(\frac{\theta}{2}\right) - \cot(2^{n-1} \theta)$

Let

$$T_n = \operatorname{cosec}(2^{n-1} \theta) = \frac{1}{\sin(2^{n-1} \theta)}$$

$$= \frac{\sin(2^{n-2} \theta)}{\sin(2^{n-1} \theta)} = \frac{\sin(2^{n-1} \theta - 2^{n-2} \theta)}{\sin(2^{n-1} \theta) \sin(2^{n-2} \theta)}$$

$$= \frac{\sin(2^{n-1} \theta) \cos(2^{n-2} \theta) - \cos(2^{n-1} \theta) \sin(2^{n-2} \theta)}{\sin(2^{n-1} \theta) \sin(2^{n-2} \theta)}$$

$$= \cot(2^{n-2} \theta) \cot(2^{n-1} \theta)$$

$$\Rightarrow \sum_{r=1}^n \cos \sec(2^{r-1} \theta) = \left[\begin{array}{l} \cot\left(\frac{\theta}{2}\right) - \cot \theta \\ + \cot(\theta) - \cot(2\theta) \\ + \\ \vdots \\ + \\ \cot(2^{n-3} \theta) - \cot(2^{n-2} \theta) \\ + \cot(2^{n-2} \theta) - \cot(2^{n-1} \theta) \end{array} \right]$$

$$= \cot\left(\frac{\theta}{2}\right) - \cot(2^{n-1} \theta)$$

28. Let $f_n(\theta) = \sum_{r=1}^n \tan\left(\frac{\theta}{2^r}\right) \sec\left(\frac{\theta}{2^{r-1}}\right)$, then compute the value of:

- (a) $f_3(2\pi)$, (b) $f_4\left(\frac{4\pi}{3}\right)$, (c) $f_5(4\pi)$ (d) $f_6(48\pi)$.

यदि $f_n(\theta) = \sum_{r=1}^n \tan\left(\frac{\theta}{2^r}\right) \sec\left(\frac{\theta}{2^{r-1}}\right)$ तो निम्नलिखित के मान ज्ञात करिये।

- (a) $f_3(2\pi)$, (b) $f_4\left(\frac{4\pi}{3}\right)$, (c) $f_5(4\pi)$ (d) $f_6(48\pi)$.

Ans. (a) -1 , (b) $2(\sqrt{3}-1)$, (c) $1-\sqrt{2}$, (d) 1

Sol. $f_n(\theta) = \sum_{r=1}^n \tan\left(\frac{\theta}{2^r}\right) \sec\left(\frac{\theta}{2^{r-1}}\right)$

$$t_r = \tan\left(\frac{\theta}{2^r}\right) \sec\left(\frac{\theta}{2^{r-1}}\right) = \frac{\sin\left(\frac{\theta}{2^r}\right)}{\cos\left(\frac{\theta}{2^r}\right) \cos\left(\frac{\theta}{2^{r-1}}\right)}$$

$$= \frac{\sin\left[\frac{\theta}{2^{r-1}} - \frac{\theta}{2^{r-2}}\right]}{\cos\left(\frac{\theta}{2^r}\right)\cos\left(\frac{\theta}{2^{r-1}}\right)} = \tan\left(\frac{\theta}{2^{r-1}}\right) - \tan\left(\frac{\theta}{2^r}\right)$$

$$f_n(\theta) = \tan\theta - \tan\frac{\theta}{2} + \tan\frac{\theta}{2} - \tan\frac{\theta}{4} + \dots + \tan\frac{\theta}{2^{n-2}} - \tan\frac{\theta}{2^{n-1}} + \tan\frac{\theta}{2^{n-1}}$$

$$= \tan\theta - \tan\left(\frac{\theta}{2^n}\right)$$

$$(a) f_3(2\pi) = \tan 2\pi - \tan\left(\frac{2\pi}{3}\right) = 0 - 1 = -1$$

$$(b) f_4\left(\frac{4\pi}{3}\right) = \tan\left(\frac{4\pi}{3}\right) - \tan\left(\frac{4\pi}{3 \cdot 16}\right) = \tan\frac{\pi}{3} - \tan\left(\frac{\pi}{12}\right)$$

$$= \sqrt{3} - (2 - \sqrt{3}) = +2\sqrt{3} - 2$$

$$(c) f_5(4\pi) = \tan 4\pi - \tan\left(\frac{4\pi}{32}\right) = -\tan\left(\frac{\pi}{8}\right) = -(\sqrt{2} - 1) = 1 - \sqrt{2}$$

$$(d) f_6(48\pi) = \tan(48\pi) - \tan\left(\frac{48\pi}{64}\right) = -\tan\left(\frac{3\pi}{4}\right) = 1$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 29 to 31

29. The minimum value of $16 \cos^4 \theta - 20 \cos^2 \theta + 5$ is :

- (A) 1 (B) $-\frac{5}{4}$ (C) 0 (D) None of these

$16 \cos^4 \theta - 20 \cos^2 \theta + 5$ का न्यूनतम मान क्या है :

- (A) 1 (B) $-\frac{5}{4}$ (C) 0 (D) इनमें से कोई नहीं

Ans. B

Sol. $y = 16 \cos^4 \theta - 20 \cos^2 \theta + 5$

$$\Rightarrow y = 4 \left[4 \cos^4 \theta - 5 \cos^2 \theta \right] + 5$$

$$\Rightarrow y = 4 \left[\left(2 \cos^2 \theta - \frac{5}{4} \right)^2 \right] + 5 - \frac{25}{4}$$

$$\therefore \cos^2 \theta \in [0, 1]$$

$$\Rightarrow 0 \leq 2 \cos^2 \theta \leq 2$$

$$\Rightarrow \frac{-5}{4} \leq \left(2 \cos^2 \theta - \frac{5}{4} \right) \leq \frac{3}{4}$$

$$\Rightarrow 0 \leq \left(2 \cos^2 \theta - \frac{5}{4} \right) \leq \frac{25}{16}$$

$$\Rightarrow 0 \leq 4 \left(2 \cos^2 \theta - \frac{5}{4} \right)^2 \leq \frac{25}{4}$$

$$\Rightarrow -\frac{5}{4} \leq y \leq 5$$

$$\Rightarrow \text{Minimum value of } y \text{ is } -\frac{5}{4}$$

30. The maximum and minimum values of $|\sin x| + \cos x$ when $x \in \mathbb{R}$ must be :

$|\sin x| + \cos x$ जहाँ $x \in \mathbb{R}$ का महत्तम व न्यूनतम मान क्या होगा?

- (A) $[0, 1]$ (B) $[-1, \sqrt{2}]$ (C) $[-\sqrt{2}, 1]$ (D) $[0, \sqrt{2}]$

Ans. B

Sol. $y = |\sin x| + \cos x$

$$y = \begin{cases} \sin x + \cos x & ; x \in [0, \pi] \\ \cos x - \sin x & ; x \in [\pi, 2\pi] \end{cases}$$

$$\because \sin x + \cos x \in [-\sqrt{2}, \sqrt{2}] \quad \forall x \in \mathbb{R}$$

$$\text{In } 0 \leq x \leq \pi; y = \sin x + \cos x$$

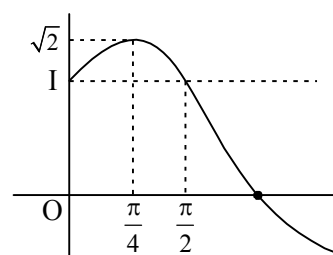
$$\Rightarrow y \in [-1, \sqrt{2}]$$

$$\text{In } \pi \leq x \leq 2\pi$$

$$y = \cos x - \sin x$$

$$\Rightarrow y \in [-1, \sqrt{2}]$$

$$y \in [-1, \sqrt{2}]$$



31. The range of the function $|\sin x| + |\cos x|$ must be :

- (A) $[-\sqrt{2}, \sqrt{2}]$ (B) $[1, \sqrt{2}]$ (C) $[0, \sqrt{2}]$ (D) None of these

$|\sin x| + |\cos x|$ फलन का परिसर होगा :

- (A) $[-\sqrt{2}, \sqrt{2}]$ (B) $[1, \sqrt{2}]$ (C) $[0, \sqrt{2}]$ (D) इनमें से कोई नहीं

Ans. B

Sol. $y = |\sin x| + |\cos x|$

$$= \begin{cases} \sin x + \cos x & ; 0 \leq x \leq \frac{\pi}{2} \\ \sin x - \cos x & ; \frac{\pi}{2} \leq x \leq \pi \\ -\sin x - \cos x & ; \pi \leq x \leq \frac{3\pi}{2} \\ -\sin x + \cos x & ; \frac{3\pi}{2} \leq x \leq 2\pi \end{cases}$$

$$\therefore \sin x + \cos x \in [-\sqrt{2}, \sqrt{2}]$$

$$\text{In } 0 \leq x \leq \frac{\pi}{2}; \sin x + \cos x \in [1, \sqrt{2}]$$

$$\text{In } \frac{\pi}{2} \leq x \leq \pi; \sin x - \cos x \in [1, \sqrt{2}]$$

$$\pi \leq x \leq \frac{3\pi}{2}; \sin x - \cos x \in [1, \sqrt{2}]$$

$$\frac{3\pi}{2} \leq x \leq 2\pi; \sin x - \cos x \in [1, \sqrt{2}]$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

32. If $\sec \theta + \tan \theta = 1$, then one root of equation, $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ is :

यदि $\sec \theta + \tan \theta = 1$, तो समीकरण $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ का एक मूल होगा :

- (A) $\tan \theta$ (B) $\sec \theta$ (C) $\cos \theta$ (D) $\sin \theta$

Ans. BC

Sol. $(ab - ac)x^2 + (bc - ab)x + (ac - ab) = 0$

$\therefore$ sum of coefficient is zero $\Rightarrow$ one root will be 1

$\therefore \sec \theta + \tan \theta = 1$ & $\sec \theta - \tan \theta = 1$

$\Rightarrow \sec \theta = 1$ & $\tan \theta = 0$

$\Rightarrow \theta = 0^\circ$

$\Rightarrow \cos \theta = 1, \sec \theta = 1, \sin \theta = 0, \tan \theta = 0$

33. If $\sin \alpha = \frac{3}{5}$, $\frac{\pi}{2} < \alpha < \pi$ and $\cos \beta = \frac{-5}{13}$, $\pi < \beta < \frac{3\pi}{2}$, then the correct statement(s) is(are) :

यदि $\sin \alpha = \frac{3}{5}$, $\frac{\pi}{2} < \alpha < \pi$ तथा $\cos \beta = \frac{-5}{13}$, $\pi < \beta < \frac{3\pi}{2}$ है, तो सत्य कथन होंगे :

(A) $\tan(\alpha - \beta) = \frac{63}{16}$ (B) $\tan(\alpha + \beta) = \frac{33}{56}$

(C) $\sin 2\alpha = \frac{-24}{25}$ (D) $\cos 2\beta = \frac{-119}{169}$

Ans. ABCD

Sol. $\sin \alpha = \frac{3}{5}$, $\frac{\pi}{2} < \alpha < \pi$

$$\Rightarrow \cos \alpha = -\sqrt{1 - \left(\frac{3}{5}\right)^2} = \sqrt{1 - \frac{9}{25}} = \frac{-4}{5}$$

Also, $\cos \beta = \frac{-5}{13}$, $\pi < \beta < \frac{3\pi}{2}$

$$\Rightarrow \sin \beta = -\sqrt{1 - \cos^2 \beta} = -\sqrt{1 - \frac{25}{169}} = \frac{-12}{13}$$

Now,

$$(A) \quad \tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} = \frac{\left(\frac{-3}{4}\right) - \left(\frac{12}{5}\right)}{1 + \left(\frac{-3}{4}\right)\left(\frac{12}{5}\right)} = \frac{\frac{-15 - 48}{20}}{\frac{20 - 36}{20}} = \frac{-63}{-16} = \frac{63}{16}$$

$$(B) \quad \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{\frac{-3}{4} + \frac{12}{5}}{1 - \left(\frac{-3}{4}\right)\left(\frac{12}{5}\right)} = \frac{\frac{-15 + 48}{20}}{\frac{20 + 36}{20}} = \frac{33}{56}$$

$$(C) \quad \sin 2\alpha = 2 \sin \alpha \cos \alpha = 2 \left(\frac{3}{5}\right)\left(\frac{-4}{5}\right) = \frac{-24}{25}$$

$$(D) \quad \cos 2\beta = 2 \cos^2 \beta - 1 = 2 \times \frac{25}{169} - 1 = \frac{50 - 169}{169} = \frac{-119}{169}$$

34. If $\sin x = \frac{1}{4}(\sqrt{5} - 1)$ then which of the following is/are correct?

यदि $\sin x = \frac{1}{4}(\sqrt{5} - 1)$ है, तो निम्न में से कौनसा/कौनसे सत्य है?

(A) $\cos 4x = \sin x$

(B) $\sin 3x = \cos 2x$

(C) $\sin 2x + \cos 3x = \frac{1}{2}$

(D) $\sin 5x + \cos 5x = 1$

Ans. ABD

Sol. $\sin x = \frac{\sqrt{5}-1}{4}$

$\Rightarrow x = 18^\circ$

(A) $\cos 4x = \cos 72^\circ = \sin 18^\circ$

(B) $\cos 2x = \cos 36^\circ = \sin (54^\circ) = \sin 3x$

(C) $\sin 36^\circ + \cos 54^\circ = \sin 36^\circ + \sin 36^\circ = 2\sin 36^\circ = 4 \times \sin 18^\circ \cos 18^\circ$

$= 4 \times \frac{\sqrt{5}-1}{4} \times \frac{\sqrt{5}+1}{4} = 1$

(D) $\sin 90^\circ + \cos 90^\circ = 1$

35. If the equation $2 \cos^2 x + \cos x - a = 0$ has solutions then a can be :

यदि समीकरण $2 \cos^2 x + \cos x - a = 0$ के हल हैं, तो a हो सकता है :

(A) $\frac{-1}{4}$

(B) $\frac{-1}{8}$,

(C) 2

(D) 5

Ans. BC

Sol. $2 \cos^2 x + \cos x - a = 0$

$\Rightarrow \cos x = \frac{-1 \pm \sqrt{1+8a}}{4}$

$\Rightarrow$ For real root $1+8a \geq 0$

$\Rightarrow a \geq \frac{-1}{8}$

(C) If $a = 2$

then $\cos x = \frac{-1 \pm \sqrt{17}}{4}$

(D) If $a = 5$

then $\cos x = \frac{-1 \pm \sqrt{41}}{4} \in (-\infty, -1) \cup (1, +\infty)$

36. $\frac{\sin^2 50^\circ - \sin^2 40^\circ}{\cos^2 50^\circ \cdot \cos^2 40^\circ}$ is equal to :

$\frac{\sin^2 50^\circ - \sin^2 40^\circ}{\cos^2 50^\circ \cdot \cos^2 40^\circ}$ बराबर होगा :

(A) $4 \tan 10^\circ \sec 10^\circ$

(B) $2 \tan 10^\circ \sec 10^\circ$

(C) $4 \tan 10^\circ (\tan 10^\circ + \cot 40^\circ)$

(D) $4 \tan 10^\circ (\tan 10^\circ + \tan 40^\circ)$

Ans. AD

Sol. $\frac{\sin^2 50^\circ - \sin^2 40^\circ}{\cos^2 50^\circ \cdot \cos^2 40^\circ} = \frac{4(\sin^2 50^\circ - \sin^2 40^\circ)}{(2 \cos 50^\circ \cdot \cos 40^\circ)^2} = \frac{4 \sin 90^\circ \cdot \sin 10^\circ}{(\cos 90^\circ + \cos 10^\circ)^2} = \frac{4 \sin 10^\circ}{\cos 10^\circ \cdot \cos 10^\circ}$
 $= 4 \tan 10^\circ \sec 10^\circ$

now (D) $\rightarrow 4 \tan 10^\circ \left[\frac{\sin 10^\circ \cos 40^\circ + \sin 40^\circ \cos 10^\circ}{\cos 10^\circ \cdot \cos 40^\circ} \right] = 4 \tan 10^\circ \left[\frac{\sin 50^\circ}{\cos 10^\circ \cdot \cos 40^\circ} \right]$
 $= 4 \tan 10^\circ \frac{1}{\cos 10^\circ} = 4 \tan 10^\circ \sec 10^\circ$

37. Which of following functions have the maximum value unity?

निम्न में से कौनसे फलन का अधिकतम मान इकाई है?

(A) $\sin^2 x - \cos^2 x$

(B) $\sqrt{\frac{6}{5}} \left(\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{3}} \cos x \right)$

(C) $\cos^6 x + \sin^6 x$

(D) $\cos^2 x + \sin^4 x$

Ans. ABCD

Sol. (A) $\sin^2 x - \cos^2 x = 1 - 2\cos^2 x = -\cos 2x$
 $\Rightarrow \sin^2 x - \cos^2 x \in [-1, 1]$

(B) Let $y = \sqrt{\frac{6}{5}} \left(\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{3}} \cos x \right)$

$\therefore \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{3}} \cos x \in \left[-\sqrt{\frac{1}{2} + \frac{1}{3}}, \sqrt{\frac{1}{2} + \frac{1}{3}} \right] \Rightarrow y \in [-1, 1]$

(C) $\cos^6 x + \sin^6 x = (\cos^2 x + \sin^2 x)^3 - 3\cos^2 x \sin^2 x (\cos^2 x + \sin^2 x)$
 $= 1 - 3\sin^2 x \cos^2 x$

$= 1 - \frac{3}{4} \sin^2 2x$

$$\Rightarrow \frac{1}{4} \leq \cos^6 x + \sin^6 x \leq 1 \quad \left\{ \because 0 \leq \sin^2 2x \leq 1 \right\}$$

$$\begin{aligned} \text{(D)} \cos^2 x + \sin^4 x &= \cos^2 x + (\sin^2 x)^2 \\ &= \cos^2 x + (1 - \cos^2 x)^2 \\ &= \cos^4 x + \cos^2 x + 1 - 2\cos^2 x \\ &= \cos^4 x - \cos^2 x + 1 \\ &= \cos^2 x (\cos^2 x - 1) + 1 \\ &= -\sin^2 x \cos^2 x + 1 \end{aligned}$$

$$= 1 - \frac{1}{4} \sin^2 2x$$

$$\because 0 \leq \sin^2 2x \leq 1$$

$$\Rightarrow -\frac{1}{4} \leq -\frac{1}{4} \sin^2 2x \leq 0$$

$$\Rightarrow \frac{3}{4} \leq \cos^2 x + \sin^4 x \leq 1$$

38. If $\cos \beta$ is geometric mean between $\sin \alpha$ and $\cos \alpha$, where $0 < \alpha, \beta < \frac{\pi}{2}$, then $\cos 2\beta =$

यदि $\sin \alpha$ तथा $\cos \alpha$ का गुणोत्तर माध्य $\cos \beta$ है, जहाँ $0 < \alpha, \beta < \frac{\pi}{2}$, तो $\cos 2\beta =$

$$\text{(A)} -2 \sin^2 \left(\frac{\pi}{4} - \alpha \right) \quad \text{(B)} -2 \cos^2 \left(\frac{\pi}{4} + \alpha \right) \quad \text{(C)} 2 \sin^2 \left(\frac{\pi}{4} + \alpha \right) \quad \text{(D)} 2 \cos^2 \left(\frac{\pi}{4} - \alpha \right)$$

Ans. AB

Sol. $\cos \beta$ is geometric mean between $\sin \alpha$ and $\cos \alpha$, where $0 < \alpha, \beta < \frac{\pi}{2}$

$$\Rightarrow \cos \beta = \sqrt{\sin \alpha \cos \alpha} \quad \Rightarrow \cos^2 \beta = \sin \alpha \cos \alpha$$

$$\Rightarrow \cos 2\beta = 2 \cos^2 \beta - 1$$

$$= 2 \sin \alpha \cos \alpha - 1$$

$$= \sin 2\alpha - 1 = -[\sin \alpha - \cos \alpha]^2$$

Now

$$\text{(A)} -2 \sin^2 \left(\frac{\pi}{4} - \alpha \right) = -2 \left[\sin \left(\frac{\pi}{4} - \alpha \right) \right]^2 = -(\cos \alpha - \sin \alpha)^2$$

$$\text{(B)} -2 \cos^2 \left(\frac{\pi}{4} + \alpha \right) = -2 \left[\cos \left(\frac{\pi}{4} + \alpha \right) \right]^2$$

$$\begin{aligned}
&= -2 \left[\frac{1}{\sqrt{2}} \cos \alpha - \frac{\sin \alpha}{\sqrt{2}} \right]^2 = - \left[\cos^2 \alpha + \sin^2 \alpha - 2 \sin \alpha \cos \alpha \right] \\
&= 2 \sin \alpha \cos \alpha - 1 \\
&= \sin 2\alpha - 1 = \cos 2\beta
\end{aligned}$$

$$(C) \ 2 \sin^2 \left(\frac{\pi}{4} + \alpha \right) = 2 \left[\frac{\cos \alpha}{\sqrt{2}} + \frac{\sin \alpha}{\sqrt{2}} \right]^2 = 1 + \sin 2\alpha$$

$$(D) \ 2 \cos^2 \left(\frac{\pi}{4} - \alpha \right) = 2 \left[\frac{\cos \alpha}{\sqrt{2}} + \frac{\sin \alpha}{\sqrt{2}} \right]^2 = 1 + \sin 2\alpha$$

39. If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then :

[JEE-2009]

यदि $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$ हो, तो :

[JEE-2009]

$$(A) \tan^2 x = \frac{2}{3}$$

$$(B) \frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$$

$$(C) \tan^2 x = \frac{1}{3}$$

$$(D) \frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$$

Ans. AB

Sol. $\frac{\tan^4 x}{2} + \frac{1}{3} = \frac{\sec^4 x}{5}$

Let $\tan^2 x = t$

$$\Rightarrow \frac{t^2}{2} + \frac{1}{3} = \frac{(1+t)^2}{5}$$

$$\Rightarrow \frac{3t^2 + 2}{6} = \frac{t^2 + 2t + 1}{5}$$

$$\Rightarrow t = \frac{2}{3}$$

$$\Rightarrow \sin^2 x = \frac{2}{5} \text{ \& } \cos^2 x = \frac{3}{5}$$

$$\Rightarrow \frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$$

40. For $0 < \theta < \frac{\pi}{2}$, the solution(s) of $\sum_{m=1}^6 \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right) \operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ is (are) : [JEE-2009]

यदि $0 < \theta < \frac{\pi}{2}$ तो समीकरण $\sum_{m=1}^6 \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right) \operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ के हल होंगे : [JEE-2009]

- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{12}$ (D) $\frac{5\pi}{12}$

Ans. CD

Sol.
$$\sqrt{2} \sum_{m=1}^6 \frac{\sin\left[\left(\theta + \frac{m\pi}{4}\right) - \left(\theta + \frac{m\pi}{4} - \frac{\pi}{4}\right)\right]}{\sin\left(\theta + \frac{m\pi}{4} - \frac{\pi}{4}\right) \sin\left(\theta + \frac{m\pi}{4}\right)} = 4\sqrt{2}$$

$$\Rightarrow \sqrt{2} \sum_{m=1}^6 \left[\cot\left(\theta + \frac{(m-1)\pi}{4}\right) - \cot\left(\theta + \frac{m\pi}{4}\right) \right] = 4\sqrt{2}$$

$$\Rightarrow \cot \theta + \tan \theta = 4.$$

$$\Rightarrow \tan \theta = 2 \pm \sqrt{3}$$

$$\Rightarrow \theta = \frac{\pi}{12} \text{ or } \frac{5\pi}{12}$$

MATCH THE COLUMN TYPE QUESTION(S)

41. If α and β satisfy the equation, $a \cos \theta + b \sin \theta = c$, then :

Column-I

- (A) the value of $\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2}$ is
- (B) the value of $\tan \frac{\alpha}{2} + \tan \frac{\beta}{2}$ is
- (C) the value of $\tan^2 \frac{\alpha}{2} + \tan^2 \frac{\beta}{2}$ is
- (D) the value of $\left(\tan \frac{\alpha}{2} - \tan \frac{\beta}{2}\right)^2$ is

Column-II

- (P) $\frac{2b}{c+a}$
- (Q) $\frac{c-a}{c+a}$
- (R) $\frac{4(b^2 + a^2 - c^2)}{(c+a)^2}$
- (S) $\frac{2(2b^2 + a^2 - c^2)}{(c+a)^2}$

यदि α एवं β समीकरण $a \cos \theta + b \sin \theta = c$ को संतुष्ट करते हो तब :

Column-I

(A) $\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2}$ का मान होगा

(B) $\tan \frac{\alpha}{2} + \tan \frac{\beta}{2}$ का मान होगा

(C) $\tan^2 \frac{\alpha}{2} + \tan^2 \frac{\beta}{2}$ का मान होगा

(D) $\left(\tan \frac{\alpha}{2} - \tan \frac{\beta}{2} \right)^2$ का मान होगा

Column-II

(P) $\frac{2b}{c+a}$

(Q) $\frac{c-a}{c+a}$

(R) $\frac{4(b^2 + a^2 - c^2)}{(c+a)^2}$

(S) $\frac{2(2b^2 + a^2 - c^2)}{(c+a)^2}$

Ans. (A) Q (B) P (C) S (D) R

Sol. $a \cos \theta + b \sin \theta = c \begin{cases} \alpha \\ \beta \end{cases}$

$$\Rightarrow a \left(\frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} \right) + \frac{2b \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} = c$$

let $\tan \frac{\theta}{2} = t$

$$a(1 - t^2) + 2bt = c(1 + t^2)$$

$$\Rightarrow c + ct^2 - a + at^2 - 2bt = 0$$

$$\Rightarrow (a+c)t^2 - 2bt + (c-a) = 0 \begin{cases} \tan \frac{\alpha}{2} \\ \tan \frac{\beta}{2} \end{cases}$$

(A) $\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2} = \frac{c-a}{a+c}$ (Q)

(B) $\tan \frac{\alpha}{2} + \tan \frac{\beta}{2} = \frac{2b}{a+c}$ (P)

(C) $\tan^2 \frac{\alpha}{2} + \tan^2 \frac{\beta}{2} = \left(\tan \frac{\alpha}{2} + \tan \frac{\beta}{2} \right)^2 - 2 \tan \frac{\alpha}{2} \tan \frac{\beta}{2}$

$$= \frac{4b^2}{(a+c)^2} - \frac{2(c-a)}{a+c}$$

$$= \frac{4b^2 - 2(c^2 - a^2)}{(a+c)^2} = \frac{2(2b^2 + a^2 - c^2)}{(c+a)^2} \quad (S)$$

$$\begin{aligned} (D) \left(\tan \frac{\alpha}{2} - \tan \frac{\beta}{2} \right)^2 &= \tan^2 \frac{\alpha}{2} + \tan^2 \frac{\beta}{2} - 2 \tan \frac{\alpha}{2} \tan \frac{\beta}{2} \\ &= \frac{4b^2 + 2a^2 - 2c^2}{(a+c)^2} - \frac{2(c-a)}{a+c} \\ &= \frac{4b^2 + 2a^2 - 2c^2 - 2c^2 + 2a^2}{(a+c)^2} \\ &= \frac{4(b^2 + a^2 - c^2)}{(a+c)^2} \quad (R) \end{aligned}$$

42. Match the following column :

Column-I

Column-II

(A) The value of $\cos \operatorname{ec} \frac{\pi}{18} - \sqrt{3} \sec \frac{\pi}{18}$ is

(P) 1

(B) $\frac{\cos 20^\circ + 8 \sin 70^\circ \sin 50^\circ \sin 10^\circ}{\sin^2 80^\circ}$ is equal to

(Q) 2

(C) Exact value of $\cos 20^\circ + 2 \sin^2 55^\circ - \sqrt{2} \sin 65^\circ$ is

(R) 3

(D) Let $f(\theta) = \frac{1}{2} + \frac{2}{3} \operatorname{cosec}^2 \theta + \frac{3}{8} \sec^2 \theta$. The least value of $f(\theta)$

(S) 4

for all permissible values of θ , is greater than

निम्न कॉलम को सुमेलित कीजिए :

Column-I

Column-II

(A) $\cos \operatorname{ec} \frac{\pi}{18} - \sqrt{3} \sec \frac{\pi}{18}$ का मान होगा

(P) 1

(B) $\frac{\cos 20^\circ + 8 \sin 70^\circ \sin 50^\circ \sin 10^\circ}{\sin^2 80^\circ}$ का मान होगा

(Q) 2

(C) $\cos 20^\circ + 2 \sin^2 55^\circ - \sqrt{2} \sin 65^\circ$ का मान होगा

(R) 3

(D) यदि $f(\theta) = \frac{1}{2} + \frac{2}{3} \operatorname{cosec}^2 \theta + \frac{3}{8} \sec^2 \theta$ तो θ के सभी संभावित मान

(S) 4

$f(\theta)$ हमेशा किस-किस से बड़ा होगा।

Ans. (A) S (B) Q (C) P (D) PQ

Sol. (A)
$$\frac{1}{\sin \pi/18} - \frac{\sqrt{3}}{\cos \pi/18} = \frac{2 \left[\frac{1}{2} \cos \frac{\pi}{18} - \frac{\sqrt{3}}{2} \sin \frac{\pi}{18} \right]}{\frac{\sin(\pi/9)}{2}}$$

$$= \frac{4 \left[\sin \frac{\pi}{6} \cos \frac{\pi}{18} - \cos \frac{\pi}{6} \sin \frac{\pi}{18} \right]}{\sin \frac{\pi}{9}} = 4$$

(B)
$$\begin{aligned} \text{Nr.} &= \cos 20^\circ + 4(\cos 20^\circ - \cos 120^\circ) \sin 10^\circ \\ &= \cos 20^\circ + 2(\sin 30^\circ - \sin 10^\circ) - 4(-1/2) \sin 10^\circ \\ &= 1 + \cos 20^\circ = 2 \cos^2 10^\circ \Rightarrow 2 \cos^2 10^\circ / \sin^2 80^\circ = 2 \end{aligned}$$

(C)
$$\begin{aligned} &\cos 20^\circ + 1 - \cos 110^\circ - \sqrt{2} \sin 65^\circ \\ &2 \sin 65^\circ \sin 45^\circ - \sqrt{2} \sin 65^\circ + 1 = 1 \quad \text{Ans] } \end{aligned}$$

(D)
$$\frac{1}{2} + \frac{2}{3} + \frac{3}{8} + \frac{2}{3} \cot^2 \theta + \frac{3}{8} \tan^2 \theta \geq \frac{37}{24} + 1 = \frac{61}{24} \quad \text{by AM-GM in equality}$$

$$\left\{ \because \frac{2}{3} \cot^2 \theta + \frac{3}{8} \tan^2 \theta \geq 2 \sqrt{\frac{2}{3} \cot^2 \theta \cdot \frac{3}{8} \tan^2 \theta} \geq 1 \right\}$$